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VAPOR SEPARATOR, METHOD FOR PRODUCING VAPOR SEPARATOR, AND OUTBOARD MOTOR

Abstract

The invention provides a vapor separator easily assembled with a fuel pump while suppressing cost increases due to increase in number of parts, and a method for producing the vapor separator. A vapor separator installed in a fuel supply system is provided with: a tank for storing liquid fuel; and a fuel pump disposed in the tank. The tank includes: upper member assembled with an upper portion of the fuel pump; and lower member located below the upper member, connected to upper member, and having a holding part for holding a lower portion of the fuel pump. Lower member of tank includes a guide part located between side wall of lower member and the holding part in plan view, the guide part serving to guide lower portion of the fuel pump toward the holding part when upper member with upper portion of fuel pump attached is assembled to lower member.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a vapor separator installed in a fuel supply system of an engine, a method for producing the same, and an outboard motor equipped with the vapor separator.

BACKGROUND

[0002] A vapor separator installed in a fuel supply system for an engine of an outboard motor to supply fuel to the engine is conventionally known. In general, a vapor separator includes a tank for storing liquid fuel and a fuel pump installed in the tank.

[0003] For example, Patent Document 1 describes a vapor separator in which a case (tank) for storing liquid fuel is composed of an upper case (upper member) and a lower case (lower member), and a high-pressure fuel pump (fuel pump) is housed in the case (tank). In the vapor separator described in Patent Document 1, the upper part of the high-pressure fuel pump (fuel pump) is supported by the upper case (upper member) of the case (tank) via rubber bushing, while the lower part of the high-pressure fuel pump (fuel pump) is supported by the lower case (lower member) via cushion rubber.

CITATION LIST

Patent Literature

[0004] Patent Document 1: JP2011-73630A

SUMMARY

[0005] When assembling the vapor separator described in Patent Document 1, for example, the upper member (upper case) of the tank and the fuel pump (high-pressure fuel pump) could be assembled to the lower member (lower case) of the tank while the fuel pump is attached to the upper member.

[0006] However, since the fuel pump is itself a heavy object, the fuel pump can pivot on the bushing as a fulcrum while suspended from the upper case (upper member). Therefore, when assembling the upper member of the tank and the fuel pump to the lower member of the tank, the position of the fuel pump is not stable, making it difficult to assemble smoothly.

[0007] To stabilize the position of the fuel pump during assembly, a bracket (pump fixing part) could be provided in addition to the bushing to immovably secure the fuel pump to the upper member of the tank, but this would increase the cost due to the increased number of parts. In view of the above circumstances, an object of at least some embodiments of the present invention is to provide a vapor separator that can be easily assembled with a fuel pump while suppressing cost increases due to an increase in the number of parts, a method for producing a vapor separator, and an outboard motor.

[0008] A vapor separator according to at least some embodiments of the invention is a vapor separator installed in a fuel supply system for supplying fuel to an engine, with a tank for storing liquid fuel and a fuel pump disposed in the tank. The tank includes: an upper member assembled with an upper portion of the fuel pump; and a lower member located below the upper member, connected to the upper member, and having a holding part for holding a lower portion of the fuel pump. The lower member of the tank includes a guide part located between a side wall of the lower

member and the holding part in plan view, the guide part serving to guide the lower portion of the fuel pump toward the holding part when the upper member with the upper portion of the fuel pump attached is assembled to the lower member.

[0009] According to at least some embodiments of the present invention, since the guide part is provided to guide the lower portion of the fuel pump toward the holding part when the upper member with the upper portion of the fuel pump attached is assembled to the lower member, the fuel pump can be easily installed.

[0010] Additionally, since the bracket (pump fixing part) to immovably secure the fuel pump to the upper member of the tank is not required, it is possible to suppress cost increases due to an increase in the number of parts.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0011] FIG. 1 is a schematic view of a vapor separator according to an embodiment.

[0012] FIG. 2 is a schematic view showing the assembly of the vapor separator shown in FIG. 1.

[0013] FIG. 3 is a cross-sectional view of the vapor separator according to an embodiment.

[0014] FIG. 4 is a bottom view of a lower portion of the fuel pump of the vapor separator according to an embodiment.

[0015] FIG. 5 is a perspective view of the fuel pump of the vapor separator and a filter element according to an embodiment.

[0016] FIG. 6 is a plan view of the lower member of the tank of the vapor separator according to an embodiment.

[0017] FIG. 7 is a partial cross-sectional perspective view of the lower member of the tank of the vapor separator according to an embodiment.

[0018] FIG. 8 is a partial cross-sectional perspective view showing the assembly of the vapor separator according to an embodiment.

[0019] FIG. 9 is a cross-sectional view of the vapor separator according to another embodiment.

[0020] FIG. 10 is a partial cross-sectional perspective view of the lower member of the tank of the vapor separator according to another embodiment.

[0021] FIG. 11 is a schematic view showing a configuration of an outboard motor according to an embodiment.

DETAILED DESCRIPTION

[0022] Embodiments of the present invention will now be described in detail with reference to the accompanying drawings. It is intended, however, that unless particularly identified, dimensions, materials, shapes, relative positions, and the like of components described in the embodiments shall be interpreted as illustrative only and not intended to limit the scope of the present invention.

[0023] FIG. 1 is a schematic view of a vapor separator according to an embodiment. FIG. 2 is a schematic view showing the assembly of the vapor separator shown in FIG. 1.

[0024] As shown in FIG. 1, in some embodiments, a vapor separator 1 includes a tank 2 for storing liquid fuel, and a fuel pump 10 disposed in the tank 2.

[0025] The vapor separator 1 is installed in a fuel supply system for an engine used in an outboard motor, for example, and is used in the middle of a fuel supply pipe for supplying fuel from a fuel tank to the engine.

[0026] In the example shown in FIG. 1, the vapor separator 1 is an open vapor separator with a vent passage (not shown) that releases fuel vapor in the tank 2 to the atmosphere to prevent the internal pressure in the tank 2 from rising.

[0027] In other embodiments, the vapor separator 1 is a closed vapor separator without a vent passage that communicates with the interior space of the tank 2. In a closed vapor separator, not

only liquid fuel but also fuel vapor is led to the inlet of the fuel pump, where the fuel vapor is liquefied by the fuel pump and sent with the liquid fuel to the engine.

[0028] The tank **2** includes a fuel inlet **3** through which fuel flows in and a fuel outlet **4** through which fuel boosted by the fuel pump **10** flows out. The tank **2** defines the interior space where liquid fuel is stored and where the fuel pump **10** is housed. The interior space of the tank **2** includes a liquid phase section **100** where liquid fuel resides and a vapor phase section **200** where fuel vapor produced by vaporization of the liquid fuel resides. The interface between the liquid phase section **100** and the vapor phase section **200** is the liquid surface L in the tank **2**. When the vapor separator **1** is in use, the fuel pump **10** is at least partially below the liquid surface L in the tank **2** and immersed in the liquid fuel (liquid phase section **100**).

[0029] The tank **2** includes an upper member **2A** and a lower member **2B** that is located below the upper member **2A** and fastened to the upper member **2A**. The upper member **2A** is a lid member that closes the upper opening of the bottomed lower member **2B**. In the embodiment shown in FIG. **1**, both the fuel inlet **3** and the fuel outlet **4** are provided in the upper member **2A**. In other embodiments, at least one of the fuel inlet **3** or the fuel outlet **4** may be provided in the lower member **2B**. The joint surface **2C** between the upper member **2A** and the lower member **2B** is provided with a sealing member (not shown) to prevent fuel leakage from inside the tank **2** through the joint surface **2C**.

[0030] The upper member **2A** and the lower member **2B** of the tank **2** have an annular protruding portion **50** and a holding part **40**, which will be described below, respectively, as supports for the fuel pump **10**.

[0031] The fuel pump **10** housed in the tank **2** with the above configuration includes a pump body **20** and an elastic cover **30** covering a lower portion of the pump body **20**.

[0032] The elastic cover **30** may be formed of any material having elasticity, for example, a polymeric material (elastomer) having elasticity.

[0033] The pump body **20** includes a pumping element **21** for boosting fuel pressure, a suction part **22** for directing fuel to the pumping element **21**, and a discharge part **24** for discharging the fuel boosted by the pumping element **21**.

[0034] The pumping element **21** of the pump body **20** can be a positive displacement pump such as a diaphragm pump or vane pump, or a non-positive displacement pump such as a volute pump, axial flow pump, or mixed flow pump.

[0035] The suction part **22** may be a suction tube that communicates with the pumping element **21** in the pump body **20**. In the exemplary embodiment shown in FIG. **1**, the suction part **22** is a suction tube composed of an elbow that includes an upstream tube section along the horizontal direction and a downstream tube section along the vertical direction, and the upstream end **23** of the upstream tube section of the suction part (suction tube) **22** is connected to an outlet tube **5** of a filter element **6**, which will be described below.

[0036] The discharge part **24** may be a discharge tube that communicates with the pumping element **21** in the pump body **20**.

[0037] In the embodiment shown in FIG. **1**, the pump body **20** has the suction part **22** at a lower portion of the pump body **20** and the discharge part **24** at an upper portion of the pump body **20**. In this case, the suction part **22** is at least partially covered by the elastic cover **30** described below.

[0038] In other embodiments, the suction part **22** of the pump body **20** is provided at an upper portion of the pump body **20** and sucks liquid fuel from the liquid phase section **100** through an extension tube. On the other hand, the discharge part **24** of the pump body **20** is provided at a lower portion of the pump body **20** and discharges the fuel to the fuel outlet **4** through an extension tube. In this case, the discharge part **24** is at least partially covered by the elastic cover **30** described below.

[0039] In some embodiments, an upper portion of the fuel pump **10** is assembled to the upper member **2A** of the tank **2**, while a lower portion (elastic cover **30**) of the fuel pump **10** is held by

the holding part **40** of the lower member **2B** of the tank **2**.

[0040] In the embodiment shown in FIG. **1**, the upper member **2A** has an annular protruding portion **50** for assembling the upper portion (discharge part **24**) of the fuel pump **10**. The upper portion (discharge part **24**) of the fuel pump **10** is assembled to the annular protruding portion **50** of the upper member **2A** via an elastic bushing **52**. The elastic bushing **52** is disposed between the upper portion (discharge part **24**) of the fuel pump **10** and the upper member **2A** (annular protruding portion **50**). The elastic bushing **52** may function as a sealing member to seal the gap between the discharge part **24** of the fuel pump **10** and the upper member **2A**. The elastic bushing **52** may be formed of any material having elasticity, for example, a polymeric material (elastomer) having elasticity.

[0041] The holding part **40** of the lower member **2B** includes a mating portion **42** capable of mating with the lower portion (elastic cover **30**) of the fuel pump **10**. The holding part **40** functions to position the lower portion of the fuel pump **10** with respect to the lower member **2B**. In the exemplary embodiment shown in FIG. **1**, the mating portion **42** of the holding part **40** is a recess that engages with a protrusion of the lower portion of the fuel pump **10**. In other embodiments, the mating portion **42** of the holding part **40** is a protrusion that engages with a recess of the lower portion of the fuel pump **10**.

[0042] When assembling the vapor separator **1** with the above configuration, as shown in FIG. **2**, the upper member **2A** and the fuel pump **10** could be assembled to the lower member **2B** of the tank **2** while the fuel pump **10** is attached to the upper member **2A** of the tank **2**.

[0043] However, since the fuel pump **10** is itself a heavy object, the fuel pump **10** can pivot on the elastic bushing **52** as a fulcrum while suspended from the upper member **2A** (see the arrow **X** in FIG. **2**). Therefore, it is necessary to stabilize the position of the fuel pump **10** when assembling the upper member **2A** of the tank **2** and the fuel pump **10** to the lower member **2B** of the tank **2**.

[0044] Therefore, in some embodiments, as shown in FIGS. **1** and **2**, the lower member **2B** of the tank **2** includes a guide part **60** to guide the lower portion of the fuel pump **10** toward the holding part **40** when the upper member **2A** with the upper portion of the fuel pump **10** attached is assembled to the lower member **2B**.

[0045] The guide part **60** disposed at the lower member **2B** stabilizes the position of the fuel pump **10**, which tends to pivot on the elastic bushing **52** as a fulcrum while suspended from the upper member **2A**, thus facilitating the assembly of the lower portion of the fuel pump **10** to the holding part **40**.

[0046] In some embodiments, the guide part **60** is located between a side wall **7** of the lower member **2B** and the holding part **40** when viewed from vertically above (i.e., in plan view).

[0047] In the embodiment shown in FIGS. **1** and **2**, a portion of the guide part **60** is located between the side wall **7** of the lower member **2B** and the holding part **40** and higher than the holding part **40**. Specifically, the guide part **60** has a guide surface **61** that approaches the holding part **40** as it extends downward, and the portion of the guide part **60** including the guide surface **61** is located between the side wall **7** of the lower member **2B** and the holding part **40** and higher than the holding part **40**.

[0048] The specific configuration of the guide part **60** is not limited, as long as the guide part **60** can guide the lower portion of the fuel pump **10** toward the holding part **40** of the lower member **2B** of the tank **2** when the fuel pump **10** suspended from the upper member **2A** is lowered toward the holding part **40** of the lower member **2B**.

[0049] In an embodiment, the guide part **60** includes a vertical rib extending along the vertical direction. The direction of extension of the vertical rib may be parallel to the vertical direction, or the angle formed by the direction of extension of the vertical rib with respect to the vertical direction may be greater than 0 degrees and not greater than 30 degrees. In other embodiments, the guide part **60** is a conical guide that extends upward from the holding part **40**.

[0050] Referring to FIGS. **3** to **8**, the specific configuration of a vapor separator **1A**, in which the

guide part **60** includes vertical ribs **60A**, will be described, and referring to FIGS. **9** and **10**, the specific configuration of a vapor separator **1B**, in which the guide part **60** is a conical guide **60B**, will be described.

[0051] FIG. **3** is a cross-sectional view of the vapor separator according to an embodiment. FIG. **4** is a bottom view of the lower portion of the fuel pump of the vapor separator according to an embodiment. FIG. **5** is a perspective view of the fuel pump of the vapor separator and a filter element according to an embodiment. FIG. **6** is a plan view of the lower member of the tank of the vapor separator according to an embodiment. FIG. **7** is a partial cross-sectional perspective view of the lower member of the tank of the vapor separator according to an embodiment. FIG. **8** is a partial cross-sectional perspective view showing the assembly of the vapor separator according to an embodiment.

[0052] FIG. **9** is a cross-sectional view of the vapor separator according to another embodiment. FIG. **10** is a partial cross-sectional perspective view of the lower member of the tank of the vapor separator according to another embodiment.

[0053] Of the elements shown in FIGS. **3** to **10**, those that are the same as the elements described above in reference to FIGS. **1** and **2** are denoted by the same reference characters, and repetitive description thereof will be omitted.

[0054] In some embodiments, as shown in FIGS. **3** and **9**, the vapor separator **1** (**1A**, **1B**) includes an elastic cover **30** covering a lower portion of the pump body **20** of the fuel pump **10**, and the holding part **40** has a mating portion **42** capable of mating with the elastic cover **30**.

[0055] In the exemplary embodiments shown in FIGS. **3** and **9**, the discharge part **24** of the fuel pump **10** is composed of a discharge tube for directing fuel boosted by the fuel pump **10** to the fuel outlet **4**. The discharge part **24** of the fuel pump **10** is assembled to the inner peripheral side of the annular protruding portion **50** of the upper member **2A** of the tank **2** via an elastic bushing **52** having a plurality of annular protruding portions **53** on its outer circumference. A pressure regulator **8** is connected to a branch tube portion **25** of the discharge part **24** of the fuel pump **10** to regulate the pressure of fuel toward the fuel outlet **4** and to return excess fuel into the tank **2**.

[0056] The vapor separator **1A** shown in FIG. **3** differs from the vapor separator **1B** shown in FIG. **9** in that the guide part **60** is vertical ribs **60A** or a conical guide **60B**. However, the vapor separator **1A** and the vapor separator **1B** share the same configuration of the lower portion (elastic cover **30**) of the fuel pump **10** and the holding part **40** of the lower member **2B** of the tank **2** as described below.

[0057] In the exemplary embodiment shown in FIGS. **4** and **5**, the elastic cover **30** has a protruding portion **32** that protrudes downward from a lower surface **31** of the elastic cover **30** and can mate with the mating portion **42** of the holding part **40**. The protruding portion **32** includes an annular portion **34**, a pair of straight portions **36** extending radially from the annular portion **34**, and a cylindrical portion **38** formed integrally with the annular portion **34**.

[0058] The annular portion **34** extends in a partial circular shape along the circumferential direction between a pair of ends **35**. Each end **35** of the annular portion **34** is connected to the cylindrical portion **38**. The cylindrical portion **38** protrudes downward from the lower surface **31** of the elastic cover **30** to cover the suction part **22** of the pump body **20**.

[0059] The pair of straight portions **36** is provided along a direction perpendicular to the direction of protrusion of the suction part **22** of the pump body **20** from the elastic cover **30**.

[0060] In the exemplary embodiment shown in FIG. **5**, the amount of protrusion of each portion of the protruding portion **32** with respect to the lower surface **31** of the elastic cover **30** increases in the order from the pair of straight portions **36** to the annular portion **34** to the cylindrical portion **38**.

[0061] In some embodiments, as shown in FIGS. **6**, **7**, and **10**, the holding part **40** includes an annular protruding portion **41** protruding from the bottom surface of the lower member **2B** of the tank **2**, and a mating portion **42** disposed on the inner peripheral side of the annular protruding

portion **41**. The annular protruding portion **41** is disposed so as to surround the lower portion (elastic cover **30**) of the fuel pump **10**. The mating portion **42** engages with the protruding portion **32** of the elastic cover **30** on the inner peripheral side of the annular protruding portion **41**.
[0062] The annular protruding portion **41** is provided only in a part of the circumferential direction, so that the suction part **22** of the fuel pump **10** can protrude radially outward from the elastic cover **30** in the circumferential range where the annular protruding portion **41** is not provided.

[0063] In the exemplary embodiment shown in FIGS. **6**, **7**, and **10**, the mating portion **42** includes a first recess **44**, a pair of second recesses **46** extending radially from the first recess **44**, and a third recess **48** connected to the first recess **44**.

[0064] The first recess **44** is defined by the inner peripheral surface of an annular wall portion **45** protruding from the bottom surface of the lower member **2B** of the tank **2** and the bottom surface of the lower member **2B**, and the annular portion **34** of the mating portion **42** of the holding part **40** is fitted into the first recess **44**. Each of the second recesses **46** is defined by the side wall surfaces of a pair of straight wall portions **47** protruding from the bottom surface of the lower member **2B** and the bottom surface of the lower member **2B**, and the straight portion **36** of the mating portion **42** of the holding part **40** is fitted into the second recess **46**. Of the total of four straight wall portions **47** (**47A**, **47B**) that define the pair of second recess **46**, the first pair of straight wall portions **47A** are connected to the corresponding ends of the annular wall portion **45**, and the second pair of straight wall portions **47B** face each other across the third recess **48**. The third recess **48** is defined by the side wall surfaces of the second pair of straight wall portions **47B** and the upper surface of a raised portion **49** that protrudes from the bottom surface of the lower member **2B**, and the cylindrical portion **38** of the mating portion **42** of the holding part **40** is fitted into it. The raised portion **49** protrudes less from the bottom surface of the lower member **2B** of the tank **2** than the other protruding portions (annular wall portion **45** and straight wall portions **47**) that form the mating portion **42**. The raised portion **49** has a circular contour corresponding to the cylindrical portion **38** in plan view.

[0065] In some embodiments, as shown in FIGS. **6** to **8**, the guide part **60** includes a plurality of vertical ribs **60A** that protrude from the inner wall of the side wall **7** of the lower member **2B** of the tank **2**. The plurality of vertical ribs **60A** are arranged radially around the holding part **40** along the vertical direction.

[0066] The amount of protrusion **d** of each vertical rib **60A** from the inner surface of the side wall **7** of the lower member **2B** of the tank **2** gradually increases downward. The upper surface of the vertical rib **60A** thus functions as a guide surface **61A** that is oblique to the vertical direction such that it approaches the holding part **40** as it extends downward.

[0067] In the exemplary embodiment shown in FIGS. **6** to **8**, when viewed from the out-of-plane direction, each vertical rib **60A** has a trapezoidal shape defined by one side corresponding to the guide surface **61A** and three sides connected to the outer peripheral surface of the annular protruding portion **41** of the holding part **40**, the bottom surface of the lower member **2B**, and the side wall **7** of the lower member **2B**, respectively.

[0068] When the upper member **2A** with the upper portion (discharge part **24**) of the fuel pump **10** attached is assembled to the lower member **2B**, the plurality of vertical ribs **60A** guides the lower portion (elastic cover **30**) of the fuel pump **10** toward the holding part **40** by the guide surfaces **61A**, as shown in FIG. **8**.

[0069] In other embodiments, as shown in FIGS. **9** and **10**, the guide part **60** is a conical guide **60B** that extends upward and radially outward from the holding part **40**.

[0070] In the exemplary embodiment shown in FIGS. **9** and **10**, the conical guide **60B** is connected to the upper surface of the annular protruding portion **41** of the holding part **40**, and the inner diameter of the conical guide **60B** gradually expands upward. The upper surface of the conical guide **60B** thus functions as a guide surface **61B** that is oblique to the vertical direction such that it approaches the holding part **40** as it extends downward.

[0071] The conical guide **60B** may be provided over the entire circumferential range occupied by the annular protruding portion **41** or, as shown in FIG. **10**, only over a part of the circumferential range occupied by the annular protruding portion **41**. Further, the conical guide **60B** may be discretely provided in multiple areas included in the circumferential range occupied by the annular protruding portion **41**.

[0072] The vapor separator **1** (**1A**, **1B**) with the above configuration can be assembled by the following procedure.

[0073] First, the upper portion (discharge part **24**) of the fuel pump **10** is assembled to the upper member **2A** of the tank **2**. In this context, the upper portion (discharge part **24**) of the fuel pump **10** may be assembled to the annular protruding portion **50** of the upper member **2A** via the elastic bushing **52**.

[0074] Then, the upper member **2A** of the tank **2** and the fuel pump **10** are lowered toward the lower member **2B** while guiding the lower portion (elastic cover **30**) of the fuel pump **10** toward the holding part **40** by the guide part **60** (**60A**, **60B**) (see FIGS. **2** and **8**). At this time, the upper member **2A** and the fuel pump **10** continue to be lowered until the lower portion of the fuel pump **10** mates with the mating portion **42** of the holding part **40**. In this way, the upper portion (discharge part **24**) of the fuel pump **10** is assembled to the upper member **2A** of the tank **2**, and the lower portion (elastic cover **30**) of the fuel pump **10** is held by the holding part **40** of the lower member **2B**.

[0075] The vapor separator **1** (**1A**, **1B**) is then obtained by fastening together the upper member **2A** and the lower member **2B** of the tank **2**.

[0076] With reference to FIG. **11**, an outboard motor including the vapor separator **1** (**1A**, **1B**) with the above configuration will be described.

[0077] FIG. **11** is a schematic view showing a configuration of an outboard motor according to an embodiment. As shown in this figure, an outboard motor **400** is mounted on a transom **402** of a hull **404**. The outboard motor **400** includes the vapor separator **1**, an engine **410** for generating power by burning fuel supplied via the vapor separator **1**, and a propeller **420** driven by the power from the engine **410**.

[0078] The engine **410** includes a cylinder block **412** and a cylinder head **414**. Fuel passing through the vapor separator **1** reaches a fuel injector **418** through a delivery pipe **416**. The fuel injector **418** injects the fuel into each cylinder of the engine **410**.

[0079] The power generated by the engine **410** is transmitted to a propeller shaft **422** of the propeller **420** via a power transmission member **430**. The power transmission member **430** includes a drive shaft **432** connected to the crankshaft (not shown) of the engine **410** and a bevel gear **434** disposed between the propeller shaft **422** and the drive shaft **432**.

[0080] The characteristic points of the vapor separator and the method for producing the vapor separator according to some embodiments described above are summarized as follows.

[0081] [1] A vapor separator (**1**; **1A**, **1B**) according to at least some embodiments of the invention is a vapor separator (**1**; **1A**, **1B**) installed in a fuel supply system for supplying fuel to an engine, with a tank (**2**) for storing liquid fuel and a fuel pump (**10**) disposed in the tank (**2**). The tank (**2**) includes: an upper member (**2A**) assembled with an upper portion of the fuel pump (**10**); and a lower member (**2B**) located below the upper member (**2A**), connected to the upper member (**2A**), and having a holding part (**40**) for holding a lower portion of the fuel pump (**10**). The lower member (**2B**) of the tank (**2**) includes a guide part (**60**; **60A**, **60B**) located between a side wall (**7**) of the lower member (**2B**) and the holding part (**40**) in plan view, the guide part serving to guide the lower portion of the fuel pump (**10**) toward the holding part (**40**) when the upper member (**2A**) with the upper portion of the fuel pump (**10**) attached is assembled to the lower member (**2B**).

[0082] With the above configuration [1], since the guide part (**60**; **60A**, **60B**) is provided to guide the lower portion of the fuel pump (**10**) toward the holding part (**40**) when the upper member (**2A**) of the tank (**2**) with the upper portion of the fuel pump (**10**) attached is assembled to the lower

member (2B), the fuel pump (10) can be easily installed.

[0083] Additionally, since the bracket (pump fixing part) to immovably secure the fuel pump (10) to the upper member (2A) of the tank (2) is not required, it is possible to suppress cost increases due to an increase in the number of parts.

[0084] [2] In some embodiments, in the above configuration [1], the guide part (60; 60A, 60B) is disposed at the lower member (2B) at least partially between the side wall (7) of the lower member (2B) and the holding part (40) and higher than the holding part (40).

[0085] With the above configuration [2], since the guide part (60; 60A, 60B) is at least partially located higher than the holding part (40), when the fuel pump (10) is lowered to the lower member (2B) with the fuel pump (10) suspended from the upper member (2A) of the tank (2), the guide part (60) is likely to come into contact with the fuel pump (10) before the holding part (40). This allows the guiding function of the guide part (60; 60A, 60B) for the fuel pump (10) to be effectively utilized.

[0086] [3] In some embodiments, in the above configuration [1] or [2], the guide part (60) includes a plurality of vertical ribs (60A) arranged radially around the holding part (40) along the vertical direction so as to protrude from the inner wall of the side wall (7) of the lower member (2B) of the tank (2). The amount of protrusion of each of the vertical ribs (60A) from the inner surface of the side wall (7) of the lower member (2B) of the tank (2) increases downward.

[0087] With the above configuration [3], by using vertical ribs (60A) protruding from the inner wall of the side wall (7) of the lower member (2B) of the tank (2), the function of the guide part (60) can be achieved at a low cost while reducing the number of parts.

[0088] Additionally, since the vertical ribs (60A) protruding from the inner wall of the side wall (7) of the lower member (2B) of the tank (2) are arranged radially around the holding part (40), the fuel pump (10) can be guided toward the holding part (40) from multiple directions. Furthermore, since the amount of protrusion (d) of each vertical rib (60A) from the inner surface of the side wall (7) of the lower member (2B) of the tank (2) increases downward, the fuel pump (10) can be effectively guided toward the holding part (40) along the upper surface (61A) of the vertical rib (60A) which is oblique to the vertical direction.

[0089] [4] In some embodiments, in the above configuration [1] or [2], the guide part (60) is a conical guide (60B) that extends upward and radially outward from the holding part (40).

[0090] With the above configuration [4], in the circumferential range where the conical guide (60B) is installed, the fuel pump (10) can be effectively guided toward the holding part (40) along the upper surface (61B) of the conical guide (60B), which is oblique to the vertical direction.

[0091] [5] In some embodiments, in any of the above configurations [1] to [4], the fuel pump (10) includes: a pump body (20); and an elastic cover (30) covering a lower portion of the pump body (20). The holding part (40) includes: an annular protruding portion (41) that protrudes from a bottom surface of the lower member (2B) so as to surround the lower portion of the fuel pump (10); and a mating portion (42) capable of mating with the elastic cover (30) on the inner peripheral side of the annular protruding portion (41).

[0092] In the above configuration [5], when the upper member (2A) of the tank (2) with the upper portion of the fuel pump (10) attached is assembled to the lower member (2B), the elastic cover (30) covering the lower portion of the pump body (20) of the fuel pump (10) and the mating portion (42) on the inner peripheral side of the annular protruding portion (41) of the holding part (40) need to be fitted together.

[0093] In this regard, as described in [1], the guide part (60; 60A, 60B) located between the side wall (7) of the lower member (2B) and the holding part (40) in plan view guides the lower portion of the fuel pump (10) toward the holding part (40), so that the elastic cover (30) of the fuel pump (10) can easily mate with the mating portion (42) of the holding part (40).

[0094] [6] In some embodiments, in any of the above configurations [1] to [5], the vapor separator (1; 1A, 1B) has an elastic bushing (52) disposed between the upper portion of the fuel pump (10)

and the upper member (2A).

[0095] If the upper portion of the fuel pump (10) is assembled to the upper member (2A) of the tank (2) via the elastic bushing (52) as described in [6], the fuel pump (10) is more likely to pivot on the elastic bushing (52) as a fulcrum. Therefore, it is necessary to stabilize the position of the fuel pump (10) when assembling the upper member (2A) of the tank (2) and the fuel pump (10) to the lower member (2B) of the tank (2).

[0096] In this regard, as described in [1], when the upper member (2A) of the tank (2) with the upper portion of the fuel pump (10) attached is assembled to the lower member (2B), the guide part (60; 60A, 60B) guides the lower portion of the fuel pump (10) toward the holding part (40). Thus, the fuel pump (10) can be easily installed while suppressing cost increases due to an increase in the number of parts.

[0097] [7] A method for producing a vapor separator according to some embodiments is a method for producing the vapor separator (1; 1A, 1B) with any of the above configurations [1] to [6], including: a step of assembling the upper portion of the fuel pump (10) to the upper member (2A) of the tank (2); a step of lowering the upper member (2A) and the fuel pump (10) toward the lower member (2B) while guiding the lower portion of the fuel pump (10) toward the holding part (40) by the guide part (60; 60A, 60B); and a step of fastening together the upper member (2A) and the lower member (2B) of the tank (2A) while the lower portion of the fuel pump (10) is held by the holding part (40).

[0098] With the above method [7], as described in [1], when the upper member (2A) of the tank (2) with the upper portion of the fuel pump (10) attached is assembled to the lower member (2B), the guide part (60) located between the side wall (7) of the lower member (2B) and the holding part (40) in plan view guides the lower portion of the fuel pump (10) toward the holding part (40), so that the fuel pump (10) can be easily installed.

[0099] Additionally, since the bracket (pump fixing part) to immovably secure the fuel pump (10) to the upper member (2A) of the tank (2) is not required, it is possible to suppress cost increases due to an increase in the number of parts.

[0100] [8] An outboard motor (400) according to some embodiments is provided with: the vapor separator (1; 1A, 1B) with any of the above configurations [1] to [6]; an engine (410) for generating power by burning fuel supplied via the vapor separator (1; 1A, 1B); and a propeller (420) driven by the power from the engine (410).

[0101] With the above configuration [8], as described in [1], when the upper member (2A) of the tank (2) with the upper portion of the fuel pump (10) attached is assembled to the lower member (2B), the guide part (60) located between the side wall (7) of the lower member (2B) and the holding part (40) in plan view guides the lower portion of the fuel pump (10) toward the holding part (40), so that the fuel pump (10) can be easily installed.

[0102] Additionally, since the bracket (pump fixing part) to immovably secure the fuel pump (10) to the upper member (2A) of the tank (2) is not required, it is possible to suppress cost increases due to an increase in the number of parts.

Claims

1. A vapor separator installed in a fuel supply system for supplying fuel to an engine, comprising: a tank for storing liquid fuel; and a fuel pump disposed in the tank, the tank including: an upper member assembled with an upper portion of the fuel pump; and a lower member located below the upper member, connected to the upper member, and having a holding part for holding a lower portion of the fuel pump, the lower member of the tank including a guide part located between a side wall of the lower member and the holding part in plan view, the guide part serving to guide the lower portion of the fuel pump toward the holding part when the upper member with the upper portion of the fuel pump attached is assembled to the lower member.

2. The vapor separator according to claim 1, wherein the guide part is disposed at the lower member at least partially between the side wall of the lower member and the holding part and higher than the holding part.
 3. The vapor separator according to claim 1, wherein the guide part includes a plurality of vertical ribs arranged radially around the holding part along a vertical direction so as to protrude from an inner wall of the side wall of the lower member of the tank, and an amount of protrusion of each of the vertical ribs from an inner surface of the side wall of the lower member of the tank increases downward.
 4. The vapor separator according to claim 1, wherein the guide part is a conical guide that extends upward and radially outward from the holding part.
 5. The vapor separator according to claim 1, wherein the fuel pump includes: a pump body; and an elastic cover covering a lower portion of the pump body, and the holding part includes: an annular protruding portion that protrudes from a bottom surface of the lower member so as to surround the lower portion of the fuel pump; and a mating portion capable of mating with the elastic cover on an inner peripheral side of the annular protruding portion.
 6. The vapor separator according to claim 1, comprising an elastic bushing disposed between the upper portion of the fuel pump and the upper member.
 7. A method for producing the vapor separator according to claim 1, comprising: a step of assembling the upper portion of the fuel pump to the upper member of the tank; a step of lowering the upper member and the fuel pump toward the lower member while guiding the lower portion of the fuel pump toward the holding part by the guide part; and a step of fastening together the upper member and the lower member of the tank while the lower portion of the fuel pump is held by the holding part.
 8. An outboard motor, comprising: the vapor separator according to claim 1; an engine for generating power by burning fuel supplied via the vapor separator; and a propeller driven by the power from the engine.
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